ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

L A B R E C O R D

Professional English Lab Integrated

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: Jashwanth S

Roll No: 26CS399

Date:

Subject Name: Professional English Lab Integrated

DEPARTMENT OF HUMANITIES & SCIENCES • AFFILIATED TO ANNA UNIVERSITY • CHENNAI

Self Introduction: Written Script and One-Page Profile

Jashwanth S | First-year B.E. Computer Science and Engineering | CSE-C

Jashwanth S

First-year CSE student aiming for a career in software and application development

Links	GitHub: https://github.com/jashlogin2008-droid Email: jashlogin80@gmail.com
Summary	I am a first-year Computer Science and Engineering student developing my foundation in programming and computational thinking. My current focus is learning core concepts clearly, improving problem-solving ability, and gradually turning ideas into useful software.
Introduction	Hi, I'm Jashwanth S, a first-year B.E. Computer Science and Engineering student (batch 2026–2030, CSE-C) at St. Joseph's College of Engineering, Chennai. I am interested in learning new technologies and I work with C and Python. My career aim is software and application development, and I want to turn ideas into useful software one project at a time.
Interests	Programming languages • Application development • Systems and core computing • 3D visuals • AI-assisted coding
Skills	C and Python; Visual Studio; Git and GitHub; AI-assisted coding and vibe coding
Tools explored	Antigravity, OpenCode
Project	CricScore: a Python concept for ball-by-ball scoring of local and college cricket matches
Languages	English (developing), Tamil, Hindi (learning)
Education	B.E. Computer Science and Engineering, 1st Year (2026–2030), St. Joseph's College of Engineering, Chennai

Impressive Personality: Biographical Note

Linus Torvalds – A Great Inspiration in Computer Software

Linus Torvalds is a famous software engineer and computer programmer. He was born on December 28, 1969, in Helsinki, Finland. He is best known as the creator of the Linux operating system and Git, which are important contributions to computer software development.

Achievements

1. Created the Linux operating system in 1991.
2. Developed Git in 2005, which helps programmers manage and share their code.
3. Encouraged open-source software development.
4. Inspired millions of programmers around the world.
5. Received the Millennium Technology Prize in 2012.

Personality and Qualities

- **Innovative Thinking:** He finds new solutions to technical problems.
- **Hard Work:** He continuously works to improve software.
- **Problem-Solving:** He focuses on solving real-world programming challenges.
- **Creativity:** He developed useful tools that changed software development.
- **Teamwork:** He supports collaboration among programmers worldwide.

Why I Admire Him

I admire Linus Torvalds because he has transformed the world of computer software through his knowledge, creativity and dedication. His contributions to Linux and Git inspire me to improve my programming skills and become a successful software developer.

My Learning from Him

His life teaches me that success comes through continuous learning, practical knowledge, patience and hard work. I hope to develop useful software applications and contribute to the technology world in the future.

Conclusion

Linus Torvalds is a true inspiration for computer science students. His achievements motivate me to explore programming, learn new technologies and become a skilled software engineer.

Summary of Lecture Notes

NPTEL: Enhancing Soft Skills and Personality (Prof. T. Ravichandran, IIT Kanpur)

Course structure: 8 weeks, approximately 40 lectures.

Lecture 1: Learning Beyond Academics

The first lecture presents learning as a broader process involving perception, self-management and communication. It connects these abilities with handling conflict, stress and everyday habits, showing why personal development matters alongside technical study.

Lecture 2: Communication in a Technology-Driven World

The second lecture discusses how communication changes with technology and email. It highlights common barriers, presentation skills, reading, and the role of human relationships in making communication effective.

Lecture 3: Developing a Growth-Oriented Mindset

The third lecture introduces mindset and distinguishes fixed thinking from growth thinking. The shoe-company example illustrates how the same situation can be interpreted differently. For a software student, a growth mindset supports learning from errors, practice and continuous improvement.

FIXED MINDSET	Abilities are treated as largely unchangeable; mistakes may feel like proof of limitation.
GROWTH MINDSET	Abilities can improve through effort, learning, feedback and persistence.

Week 1: remaining discussion; Week 2 follows below

Learning mindsets form early. A belief like “I’m not smart enough” limits a person, but it isn’t a fact about their ability. The suggested “secrets” are to behave like a baby (open and unafraid), put up with critical people, reject negative stereotypes and stereotype threat, and look for a sense of belonging. The Hawthorne effect is offered as evidence that belonging and attention motivate people more than incentives do.

Week 2: Time Management, Procrastination and Assertiveness

Time can’t really be managed. What we manage is our habits and our perception of it. Monochronic people stick to rigid schedules, while polychronic people stay flexible. Others tend to judge our personality by our punctuality and by how we behave under deadline pressure. To control time, log it, look for “time leakage” (interruptions, indecision, poor organisation, social media, unmanaged email) and make room for productive work. Working at your biological peak time helps. Covey’s urgent/important quadrants and the 80/20 Pareto principle help with prioritising, and “time stealers” such as waiting and travel can be turned into “time gifters.”

The lectures trace procrastination to Parkinson’s law, fear, laziness, perfectionism and addiction, along with the plain feeling of being overwhelmed. The remedies are:

- Eat the elephant bite by bite.
- Do the “frog” (the hardest task) first.
- Use the Kaizen one-minute principle.
- Keep a clean workspace.
- Set goals you actually want and declare them publicly.
- Reward yourself.
- Use the Zeigarnik effect: once you start, you feel compelled to finish.

The week ends with a first look at assertiveness, or not saying “yes” when you mean “no.”

Week 3: Assertiveness, Anger and Positive Thinking

The lectures compare passive, assertive and aggressive people on body language, listening, feelings and outcomes. Assertiveness means stating your rights without violating anyone else's, and it ideally ends in a win-win. Practical steps include checking your relationships, dropping self-debilitating thoughts, avoiding discouraging people, following Stanley and Nancy's commandments and learning four ways to say no.

On anger, Aristotle's point is that the hard part is being angry at the right person, to the right degree, at the right time. Techniques include postponing your reaction, counting, deep breathing, physical exertion, Lincoln's unsent letter, silence and channelling anger into creative work. The lectures then turn to positive thinking through the story of two hospital patients and a "window" that faced a blank wall. Changing negative thinking takes desire and determination, introspection and discarding limitations. It also means building a positive self-image through repetition and visualisation, and bringing body, mind and world into sync.

Week 4: People Skills and Being Attractive

Forty traits make others dislike you, and their opposites make people like you. Examples include egotism, aggression, pessimism, fault-finding, gossiping, false promises, moodiness and favouritism, set against selflessness, optimism, giving credit, politeness, trustworthiness, honesty and humility. For dealing with gossip, Socrates' three-filter test (truth, goodness, utility) is suggested.

Attractiveness has physical, mental, emotional and spiritual layers. Physically, it means neat footwear, a fit body, good diet and sleep, and a signature personality. Mentally, it means intellectual charm and a touch of mystery. Emotionally, it means remembering names, valuing yourself and being yourself. Spiritually, it comes down to mindfulness, deeper connection, helping without expectation and "paying it forward."

Week 5: English Skills and Common Errors

English is the global lingua franca, and avoiding common errors is presented as part of soft skills. Using Sonia Sharma's Common Errors in English, the lectures work through repeated self-scored exercises and explanations. Topics include confused word pairs (resume/résumé, principal/principle, alone/lonely), Indianisms (passed out, joint family, cousin brother) and subject-verb agreement (collective, uncountable and plural-form nouns). The lectures also cover prepositions (since/for, at/in, by/on), articles, collocations and word choices like weather/climate. Twain and Branson are quoted on learning by exposure rather than by memorising rules.

Week 6: Humour and Money

Humour reduces stress, improves health and helps communication. It works at three levels: a smile, hearty laughter, and sarcasm, which is best avoided. Adults laugh far less than children (about 4 times a day versus 300). At work, humour resolves conflict, boosts creativity and productivity, and lets you speak the truth lightly. Suggestions include a joke board, a humour club and a humour hall.

The focus then turns to money and personality. How you treat rich and poor people, how you borrow and lend, and how you handle trust all reveal emotional maturity and integrity. Wealth-building advice includes tracking income and expenses, separating need from greed, avoiding impulse buys, and becoming indispensable through devoted, excellent work, with Michelangelo, Martin Luther King and Lincoln as examples. Real wealth also includes things money can't buy.

Week 7: Health and Love

Health affects personality, first impressions and employability, since healthy people are seen as disciplined and reliable. Managing it rests on:

- Regular exercise (morning is best, with tips for sticking to it).
- Diet: regular meals, a heavy breakfast and a light early dinner, fruits and vegetables, about 2 litres of water, and no white foods or soft drinks.
- Sleep: 7-8 hours, a good sleep environment, and avoiding gadgets, late heavy meals and alarms.

Love is defined broadly, drawing on Shakespeare, D. H. Lawrence, Gibran, Lovelace and Corinthians: real love is constant, free, patient and unconditional. Managing love means reading body language and timing, letting relationships mature through stages, handling rejection gracefully and keeping romance separate from work.

Week 8: Ethics, Etiquette, Mind, Memory and Environment

Values, ethics and etiquette make up character (“when character is lost, everything is lost”). Professional ethics combine written codes with society’s moral expectations. Poor etiquette and business etiquette are covered in detail: introductions, not interrupting, confidentiality, not criticising your company, and praising in public but reprimanding in private.

The mind is powerful but scattered, and it can be trained through auto-suggestion, positive self-talk and effective thinking (gather facts, sleep on it, trust your gut). Memory techniques include association through absurd images, remembering names and faces, number-image links, memory aids and mental exercise. The final lecture urges care for the environment, with practical steps (save water and electricity, avoid plastic, use public transport, plant trees) and a shift from anthropocentric to eco-centric thinking. It closes with Gandhi’s line that the earth provides enough for everyone’s need but not everyone’s greed.

SWOC Presentation

Strategic Career Roadmap: SWOC Analysis (slides)

Personal SWOC Analysis

Strategic Career Roadmap: path into the Software / IT field

Candidate: Jashwanth S | 26CS399 | 1st Year B.E. CSE | CSE-C

Goal: keep learning, improve consistently and prepare for a future in the Software / IT field

Executive Summary & SWOC Vision

1 Strength

Interested in learning new technologies.

2 Weakness

Working to improve decision-making, concentration and communication.

3 Opportunity

Programming, AI, software development and practical projects.

SWOC Matrix at a Glance

Strengths
Internal

Interest in learning new technologies.

Weaknesses
Internal

Decision-making, distraction, time and focus, communication, and social-emotional areas for growth.

Opportunities
External

Programming, Artificial Intelligence, software development and building projects.

Challenges
External

Keeping up with changing technology and developing strong technical skills.

Strengths

1	**Interest in New Technologies**	Keen to learn new technologies and see how they support a future in Software / IT.
2	**Interest in Programming**	Building a foundation in programming concepts and practice.
3	**Willingness to Learn**	Ready to learn the fundamentals and improve step by step.
4	**Early Project Exposure**	CricScore, a Python scoring concept, shows my interest in building useful software.

Weaknesses

1. **Decision-Making**
   Sometimes find it difficult to make quick decisions.
2. **Distraction**
   I feel distracted many times.
3. **Time & Focus**
   Need to strengthen planning, time use and sustained focus.
4. **Communication**
   Professional interaction and communication are areas to develop.
5. **Emotional & Social**
   Personal and social areas I want to handle with greater confidence.

Opportunities

1. **Programming**
   Build a foundation in programming concepts and practice.
2. **Artificial Intelligence**
   Explore AI concepts and their applications.
3. **Software Development**
   Learn how software is designed and developed.
4. **Practical Projects**
   Apply learning by creating projects.

Challenges

1. **Rapidly Changing Technology**
   Keeping up with new tools, technologies and developments in the field.

2. **Developing Technical Skills**
   Building strong technical skills through regular learning and practice.

3. **Strong Competition**
   Standing out in a competitive software field.

4. **Strong Fundamentals**
   Avoiding gaps in fundamentals if learning is rushed.

Action Plan: Closing the Gaps

1. **Better Decisions**
   Compare options, set a reasonable time limit and make a considered choice.

2. **Strengthen Focus**
   Use a simple daily task list and work in focused study sessions.

3. **Build Communication**
   Practise clear speaking, active listening and professional interaction.

Action Plan: Using Opportunities

1 Learn Programming

Build a foundation in programming concepts and practise regularly.

2 Explore AI

Understand introductory AI concepts and possible applications.

3 Develop Software Skills

Learn development concepts through guided practice.

4 Build Projects

Apply learning through small, practical projects.

4-Year Learning Roadmap

YEAR	CORE FOCUS	PROGRAMMING	AI & PROJECTS	EXPERIENCE
FIRST	CSE fundamentals	Programming basics	AI foundations	Learning habits
SECOND	Data structures	Problem solving	Introductory projects	Project practice
THIRD	Advanced CSE topics	Applied development	AI-related projects	Internships / experience
FINAL	Integration & revision	Portfolio-ready work	Refined projects	Career readiness

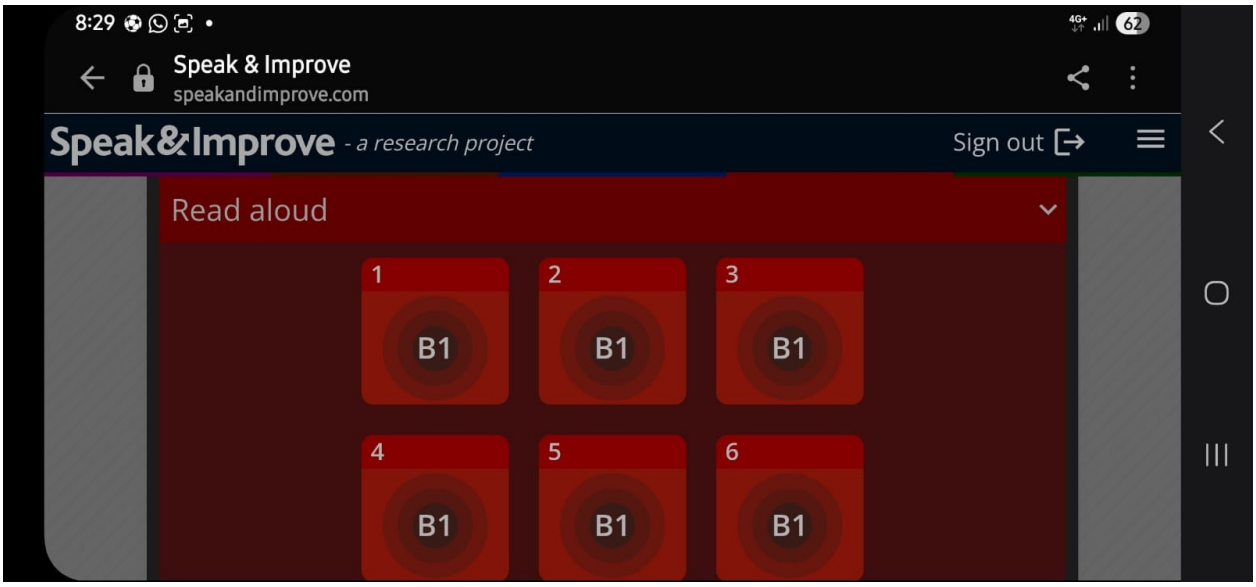
Conclusion & Key Takeaways

1. **Keep Exploring**
   My interest in new technologies can support continuous learning.
2. **Work on Growth Areas**
   Improving focus, decision-making and communication can strengthen my development.
3. **Build a Career in IT**
   Programming, AI, software development and projects support my Software / IT goal.

Thank you

Write and Improve / Speak and Improve

Speak and Improve: Read aloud tasks (screenshot)



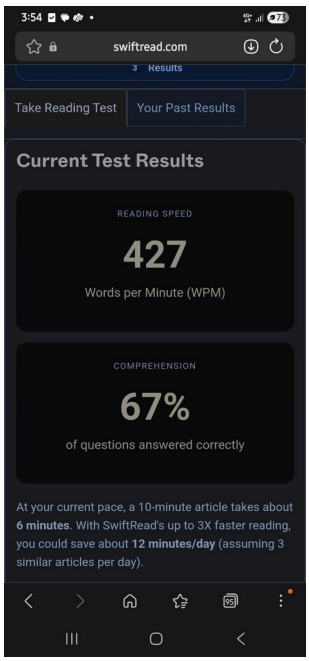
All six Read aloud tasks shown in the screenshot (tasks 1 to 6) are at level B1.

Write and Improve / Speak and Improve

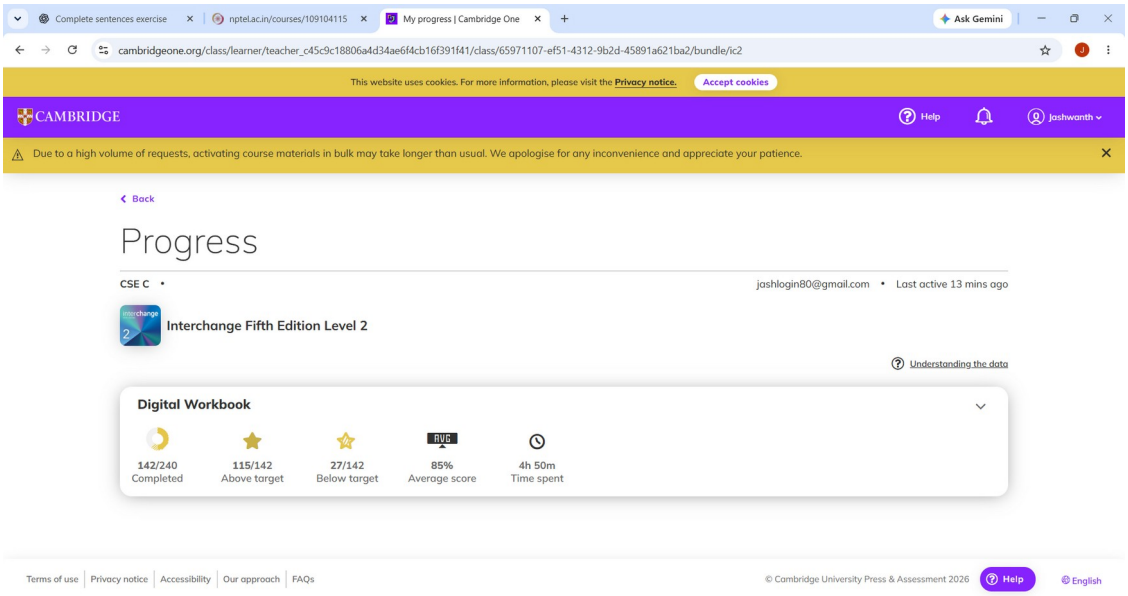
Reading speed test and Cambridge One progress

Activity	Result
SwiftRead reading test	427 words per minute; 67% comprehension
Cambridge One: Interchange Fifth Edition Level 2 (Digital Workbook)	142/240 completed; 85% average score; 4 h 50 min spent

SwiftRead reading test result

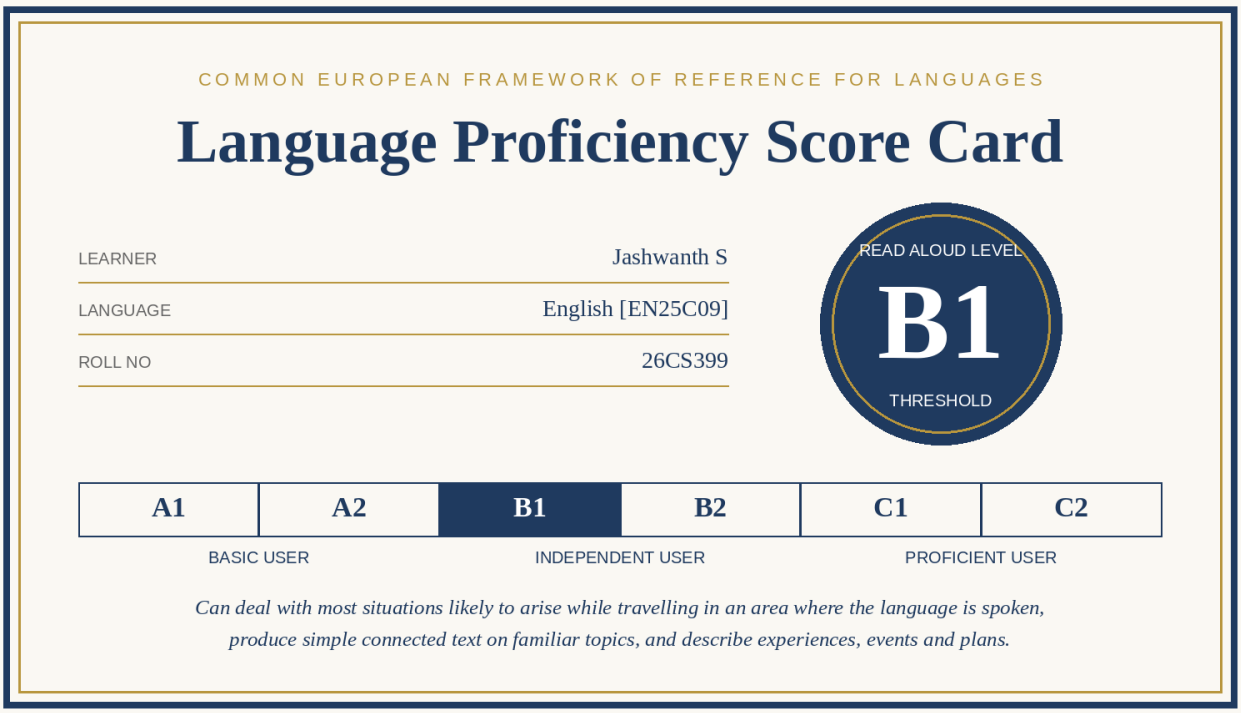


Cambridge One progress (Interchange Level 2, CSE C)



Write and Improve / Speak and Improve

CEFR score card



COMMON EUROPEAN FRAMEWORK OF REFERENCE FOR LANGUAGES

Language Proficiency Score Card

LEARNER Jashwanth S

LANGUAGE English [EN25C09]

ROLL NO 26CS399

READ ALOUD LEVEL

B1

THRESHOLD

A1	A2	B1	B2	C1	C2

BASIC USER INDEPENDENT USER PROFICIENT USER

Can deal with most situations likely to arise while travelling in an area where the language is spoken, produce simple connected text on familiar topics, and describe experiences, events and plans.

My Speak and Improve Read aloud level is B1 (Intermediate).

Professional English Practice Worksheets – Solved

Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated) | Worksheet 1: Model Exam I (Parts A, B, C), followed by the Vocabulary Worksheet

Worksheet 1 (EN25C09 Model Exam I) – Part A (15 × 1 = 15 Marks)

1. Answer: monitors

2. Answer: recorded

3. Answer: stronger

4. Answer: must

5. Answer: can

6. Answer: The laboratory developed a new testing device that measures material strength with high precision.

7. Answer: rises

8. Answer: would not have failed

9. Answer: The mechanics installed dual damper springs to isolate chassis resonance.

10. Answer: optimizing

11. Answer: It was reported that 50 mL of hydrochloric acid was poured into the titration flask.

12. Answer: smoothly

13. Answer: The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.

14. Answer: The laboratory purchased several new pieces of equipment for conducting material testing.

15. Answer: worse

Worksheet 1 – Part B (10 × 2 = 20 Marks)

1. Answer: The engineer calibrated the pressure sensor and verified its accuracy.

2. Answer: Portability: A laptop is more portable than a desktop computer. Screen size: A desktop computer generally has a larger screen than a laptop.

3. Answer: The engineer designed a new cooling system that reduces the temperature of industrial machines.

4. Answer: (a) If the battery voltage drops below the required level, the system will issue a warning. (b) If the technician had checked the connections earlier, the problem would have been detected.

5. Answer: It was reported that the solution was heated to 80°C before testing.

6. Answer: The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.

7. Answer: (a) The engineer received much feedback on the new design. (b) The students received several useful pieces of advice from their teacher.

8. Answer: (a) The technician wears safety gloves to protect his hands. (b) The cooling fan operates continuously to prevent the motor from overheating.

9. Answer: (a) efficiently (b) carefully

10. Answer: (a) can (b) must not

Worksheet 1 – Part C (5 × 13 = 65 Marks)

Question 26 (a) Safety instructions for safe and responsible use of the internet

1. Use strong and unique passwords for every account.
2. Enable two-factor authentication whenever possible.
3. Do not share your passwords, OTPs or other confidential information with anyone.
4. Keep your operating system, browser and security software updated.

5. Do not open suspicious links or unknown email attachments.
6. Use secure websites and avoid entering sensitive information on untrusted networks.
7. Check website addresses carefully before entering login or payment details.
8. Report suspicious activity and cyber threats to the appropriate authority.

Question 27 (a) Descriptive essay: A traditional art form in my region and how modern initiatives can preserve it

Selected Question: 27.a

Topic: Kolam – A Traditional Art Form of Tamil Nadu

Introduction:

Kolam is a traditional art form that is widely practised in Tamil Nadu. It is drawn using rice flour or other natural materials in the form of geometric and floral patterns, mostly in front of houses early in the morning. In my home town too, almost every house has a kolam at the entrance, and it is one of the first things people notice when they visit.

Body Paragraph 1:

Kolam has been passed down from one generation to another and is closely connected with our everyday life, festivals and hospitality. Traditionally, women have played an important role in keeping this art alive by teaching patterns and techniques to their daughters and other family members. During festivals like Pongal, people draw large and colourful kolams. Apart from its beauty, kolam also shows creativity, mathematical thinking, symmetry and patience, because the dots and lines have to be arranged very carefully.

Body Paragraph 2:

Modern initiatives can help preserve kolam so that it does not become a forgotten tradition. Schools and colleges can conduct workshops and competitions where students learn traditional designs from experienced artists. Cultural organisations can record different regional styles through photographs, videos and digital archives. Social media can also be used in a responsible way to introduce young people to the history and meaning of kolam. Artists can create new designs while still keeping the traditional methods and symbols. Exhibitions, public art programmes and festivals give young artists a chance to practise and display their work. It is equally important to support traditional artists through training and fair payment.

Conclusion:

By combining traditional knowledge with modern educational and digital initiatives, kolam can remain a living cultural practice rather than just a memory of the past. As students, we should also take an interest in learning it, so that we can pass this beautiful art on to the next generation.

Question 28 (a) Formal letter of request to the Librarian

Selected Question: 28.a

Sender's Details / Date:

From
Jashwanth S
I Year B.E., Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119.
Date: 21 October 2026

Recipient's Details:

To
The Librarian
St. Joseph's College of Engineering

OMR, Chennai – 600 119.

Subject:

Request for extension of library working hours during the semester examinations – reg.

Salutation:

Respected Sir/Madam,

Content / Body:

I am writing to request permission to extend the working hours of our college library during the semester examination period. Extended library hours would give students more time to prepare for the examinations, revise course materials, refer to technical books and complete their academic work. Many students need a quiet and well-equipped place for evening study, especially during the examination period.

A good number of students are expected to use the library during the extended hours. The additional time can be managed by maintaining proper discipline, making sure there is enough lighting and security, and assigning staff members according to their availability. Students can be instructed to maintain silence, handle the books carefully and follow all the library rules.

I kindly request you to consider this proposal and permit extended library hours during the semester examinations. This arrangement would be a great help to all of us in our academic preparation.

Sign-off:

Thanking you,
Yours faithfully,
Jashwanth S
(Roll No.: 26CS399)

Question 29 (a) Analytical essay: AI-based assessment versus traditional teacher-based assessment

Selected Question: 29.a

Criteria 1 vs. Criteria 2:

AI-Based Assessment vs. Traditional Teacher-Based Assessment – compared on assessment accuracy, feedback time, fairness and the role of human judgement.

Pros & Cons Breakdown:

AI-based assessment – Pros: evaluates a large number of answers quickly, gives immediate feedback, applies the same criteria to everyone. Cons: may not correctly judge unusual but valid methods in open-ended tasks; biased data or poorly designed algorithms can lead to unfair results.

Teacher-based assessment – Pros: understands the reasoning, creativity and individual situation of a student; feedback is more detailed and personal. Cons: takes more time because teachers must correct many papers; human judgement can also vary.

Drafting Notes:

Artificial Intelligence-based assessment and traditional teacher-based assessment both have important roles in engineering education. AI-based assessment can evaluate a large number of answers quickly and give immediate feedback. It also applies the same programmed criteria to every student, which can improve uniformity. On the other hand, traditional assessment allows teachers to understand the reasoning, creativity and individual circumstances behind a student's answer.

Accuracy is a major advantage of AI when the questions have objective answers and clear evaluation rules. However, AI may not always judge unusual but valid approaches correctly, especially in open-ended engineering tasks. Teachers can understand such answers using their professional and academic judgement. AI can give feedback

very quickly, while teacher feedback may take longer because teachers have to evaluate many students. Still, feedback from a teacher is usually more detailed and personal.

Fairness is another important point. AI can reduce some kinds of inconsistency, but biased data, poorly designed algorithms or unsuitable evaluation criteria can also lead to unfair results. Teacher-based assessment is not automatically fair either, because human judgement can also vary. So neither method is perfect.

In my opinion, the best approach is to combine both methods. AI can handle routine assessment and give quick preliminary feedback, while teachers can review important or complex work and make the final academic decisions. Such a balanced system can improve efficiency without removing the essential role of human judgement in engineering education.

Question 30 (a) Personal email: First semester in engineering

Selected Question: 30.a

To:

My School Friend

Subject Line:

My First Semester in Engineering

Salutation:

Hi Friend,

Opening / Check-in:

I hope you are doing well. I wanted to share my experience during the first semester of my engineering course. College life has been very different from school, but I am slowly getting comfortable with the new environment.

Core Experience / Details:

I have made several new friends in my class. We help one another with assignments, lab work and difficult topics. Classroom activities have also been interesting because we get chances to take part in discussions, practical sessions and technical activities. The lab classes have helped me understand how the theory we learn is actually used in practice.

Adjusting to college life took some time. I had to become more responsible about managing my timetable, completing assignments, attending classes and preparing for exams. I am still working on improving my time management.

Right now I am mainly focusing on the upcoming semester examinations. I plan to revise every subject properly, practise the important questions and really understand the topics instead of just memorising answers. I hope the next semester brings new subjects, better skills and more chances to learn.

Closing:

Write back and tell me how things are going with you.

Best wishes,
Your Friend

Worksheet 2: Model-I Practice Worksheet

The following answers are reproduced from the supplied reference. Personal names, experiences, dates and budgets in writing examples belong to that source; they are not verified facts about Jashwanth.

Worksheet 2 (Model – I Practice Worksheet) – Exercises 1 to 15

Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. recognize
2. requires
3. transmit
4. agrees
5. form

Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

Exercise 4: Modals of Obligation and Necessity

1. must
2. must
3. have to
4. must
5. has to

Exercise 5: Modals of Capability and Possibility

1. can
2. may
3. can
4. can
5. can

Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.
2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuum chamber.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the robotics lab.
4. This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remote sensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessment yesterday.

Exercise 7: If-Clauses (Type 1)

1. exceeds
2. fails
3. engages
4. rise
5. registers

Exercise 8: If-Clauses (Type 3)

1. had started
2. had lubricated
3. would have been prevented
4. would have tripped
5. had remained

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. redesigning
2. replacing
3. reducing
4. cleaning
5. filtering

Exercise 11: Impersonal Passive Voice

1. 50 mL of hydrochloric acid was poured into the titration flask.
2. The robotic arm was tested under a 50 kg dynamic payload.
3. The optical pyrometer was calibrated prior to ignition.
4. Severe cavitation damage was observed along the impeller blades.
5. Volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

1. precisely
2. periodically
3. rapidly
4. firmly
5. smoothly

Exercise 13: Reported Speech

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether he had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

1. Incorrect: "machineries" and "equipments". Correct: The procurement officer purchased several new pieces of machinery and equipment for the workshop.
2. Incorrect: "datas". Correct: The laboratory generated a huge amount of test data during the wind tunnel trials.
3. Incorrect: "softwares". Correct: We need three more software licenses to complete the finite element analysis on the model.
4. Incorrect: "informations". Correct: The field report provided a lot of valuable information regarding the soil composition.
5. Incorrect: "apparatuses" and "glasswares". Correct: All chemical apparatus and glassware must be washed thoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

1. worse
2. better
3. farther
4. less
5. further

Worksheet 2 – Part B (Exercises 1 to 10)

Exercise 1: Simple Past Tense Conversion

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmental dashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before the morning shift.

4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into a secure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulation breakdown.

Exercise 2: Comparative Sentences

1. Cloud storage has a lower initial deployment cost than on-premise physical servers because no hardware has to be bought. It is also more elastic, since its capacity can be increased within minutes.
2. Fiber-optic cables offer a much higher data transfer bandwidth than copper twisted-pair cables. They are also less susceptible to electromagnetic interference than copper cables.
3. Electric vehicles produce lower tailpipe emissions than internal combustion engine vehicles, as they release none at all. However, EVs generally have a higher initial purchase price than ICE vehicles.
4. Hydraulic actuators produce a greater raw force output than electromechanical actuators. They also carry a higher risk of fluid leakage, because electromechanical actuators use no fluid.
5. Relational (SQL) databases have a more rigid schema than document (NoSQL) databases. Document databases usually provide higher horizontal query throughput than relational databases.

Exercise 3: Relative Clauses

1. The algorithm that analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized the emergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator, where high-voltage insulator puncture tests are conducted.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoor ports.
5. The supervisory system, which utilizes an artificial neural network, predicts transformer insulation life based on dissolved gas concentrations.

Exercise 4: If-Clauses

1. (a) If the operating temperature exceeds 85°C, the cooling system will switch on automatically.

   (b) If the operator had pressed the emergency shutdown switch immediately, the accident would have been avoided.
2. (a) If the supply voltage drops below the 200V threshold, the backup power supply will take over.

   (b) If the network administrator had enabled multi-factor authentication, the unauthorized access would have been prevented.
3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the ventilation fans will start and an alarm will sound.

   (b) If the mechanical team had checked the bearing lubrication schedule, the bearing would not have overheated.
4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, it will stop immediately.

   (b) If the thermal sensor had registered the runaway reaction in time, the explosion would have been prevented.
5. (a) If the software receives an unauthenticated API token, it will reject the request and return an error message.

   (b) If the civil inspectors had reviewed the core sample density tests, they would have detected the weak foundation before construction.

Exercise 5: Impersonal Passive Voice

1. The crude extract was distilled at a constant temperature of 60°C.

2. The cross-section of the weld seam was polished with fine diamond paste.
3. Fifteen cast-iron pump casings were rejected due to internal porosities.
4. The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.
5. The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether he had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had received clearance.
3. The IT security lead enquired of the developer why he had disabled the firewall rule on the database server.
4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricane winds.
5. The research guide asked the postgraduate scholar how many sample trials he had conducted under cryogenic conditions.

Exercise 7: Countable vs. Uncountable Nouns

1. (a) We need to install several new pieces of software and machinery in the CAD laboratory.
   (b) The team collected plenty of useful information from the field test.
2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antenna lab.
   (b) High-speed train operations require advanced transport infrastructure and dedicated electrical power.
3. (a) The environmental sensor gathers extensive atmospheric data and pollution indices every minute.
   (b) The construction crew unloaded tons of sand, cement, and steel debris at the worksite.
4. (a) All electrical hardware and wiring must comply with local safety legislation.
   (b) The data scientist discovered several interesting insights after running the machine learning clustering.
5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises.
   (b) The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose and Gerund Constructions

1. (a) A pressure relief valve is installed to vent excess boiler steam.
   (b) The automated script runs every midnight to archive legacy server logs.
2. (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature.
   (b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along the girder.
3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.
   (b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.
4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.
   (b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5. (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against sudden voltage sags.
   (b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner

1. (a) automatically
   (b) flawlessly
2. (a) accurately

(b) instantly

3. (a) continuously
   (b) cautiously
4. (a) consistently
   (b) catastrophically
5. (a) periodically
   (b) smoothly

Exercise 10: Modals

1. (a) can
   (b) must not
2. (a) must
   (b) may
3. (a) must not
   (b) could
4. (a) should
   (b) must
5. (a) can
   (b) must not

Worksheet 2 – Topic 1: Writing Imperative Safety Instructions

Question 1: Chemistry and Materials Testing Laboratory

1. Wear a lab coat, safety goggles and nitrile gloves at all times inside the laboratory.
2. Handle concentrated acids and alkalis only inside the fume hood.
3. Add acid to water slowly; never add water to acid.
4. Do not smell, taste or touch any chemical directly.
5. Label every container clearly, and never return unused chemicals to the stock bottle.
6. Clean all glassware and testing equipment after each experiment to avoid contamination.
7. Keep food, drinks and open flames away from the work area.
8. Report every spill, burn or breakage to the instructor at once, and use the eyewash station or safety shower in an emergency.

Question 2: High-Voltage Electrical Machines and Drives Laboratory

1. Make sure the main supply is switched off before making or changing any connection.
2. Wear insulated rubber gloves and safety shoes, and stand on an insulated rubber mat near live panels.
3. Never touch exposed terminals, bare wires or rotating parts while the machine is running.
4. Get the circuit connections checked by the instructor before switching on the supply.
5. Know the location of the circuit breakers, fuses and emergency stop button, and keep them in working condition.
6. Remove rings, watches and metal jewellery, and avoid loose clothing, to prevent arc flash burns and accidental short circuits.
7. Keep water and wet hands away from electrical equipment, and use only insulated tools.
8. Increase the motor load gradually, and switch off the supply and discharge the capacitors before disconnecting the circuit.

Question 3: Advanced Manufacturing and CNC Machining Workshop

1. Wear safety glasses, safety shoes and close-fitting clothes, and tie back long hair.
2. Keep the machine door and guard closed while the spindle is running.
3. Do not touch the spindle, cutter or workpiece until the machine has stopped completely.
4. Clamp the workpiece and tool holder firmly before starting the program.
5. Use a brush or chip hook, and heat-resistant gloves, to remove hot metal chips, and only after stopping the machine.
6. Wipe up coolant spills immediately, and keep the floor dry to prevent slipping.
7. Run the program in simulation mode first, and keep your hand near the emergency stop button during the first cut.
8. Never leave the machine unattended or operate it without the instructor's permission.

Worksheet 2 – Topic 2: Free Writing – Festivals and Cultural Celebrations

Question 1: A traditional harvest festival and eco-friendly practices

Pongal: The Harvest Festival of Tamil Nadu and How Eco-Friendly Practices Can Protect Its Spirit

Pongal is the most important harvest festival of Tamil Nadu. It is celebrated in mid-January to thank the Sun God, the rain and the cattle for a good harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal and Kaanum Pongal.

On Bhogi, families clean their homes and throw away old things. On Thai Pongal, they draw colourful kolams at the entrance and cook fresh rice and milk in a new clay pot in the open yard. When the milk boils over, everyone shouts "Pongalo Pongal!" The pot is decorated with turmeric plants and sugarcane. The festive meal includes sakkarai pongal, venn pongal, sambar, vadai and payasam, served on banana leaves. On Mattu Pongal, farmers decorate and worship their cattle, and Kaanum Pongal is a day for family visits and community games.

Modern eco-friendly practices can preserve the core spirit of the festival. Instead of burning plastic and old tyres on Bhogi, communities can burn only dry leaves or turn them into compost, which reduces smoke and air pollution. Clay pots, banana leaves and areca-leaf plates can replace plastic and thermocol, and kolams can be drawn with natural rice flour. Villages can organise tree-planting drives, rainwater harvesting and cloth-bag campaigns during the festival, and use solar lamps instead of excess electric decorations.

In this way, Pongal can remain joyful and traditional while staying kind to nature, and the next generation will learn that celebration and conservation can go together.

Question 2: A temple car festival and sustainable measures

The Temple Car Festival: Faith, Food and a Cleaner Celebration

In my hometown, as in many towns of Tamil Nadu, the temple car festival, or Ther Thiruvizha, is the biggest event of the year. Thousands of devotees gather in the streets to watch the deity pulled in a huge, richly decorated wooden chariot, called the ther.

The festival begins with flag hoisting and several days of processions. On the car day, devotees of all ages hold thick ropes and pull the heavy chariot together through the four main streets, chanting prayers while drums and nadaswaram fill the air. This shared effort is a strong symbol of unity, because people from every community take part. Women draw kolams on the road, and families offer coconuts, bananas and flowers to the deity. Volunteers give free food such as pongal, puliyodharai, curd rice and sweet payasam as prasadam, served on leaves.

However, such a large crowd also creates problems. Single-use plastic cups, plates and bags litter the streets, and loudspeakers cause noise pollution. Eco-friendly measures can solve these problems. Organisers can ban single-use plastic and provide steel tumblers, banana leaves and paper bags. Dustbins and waste-sorting points can be placed along the route, and volunteers can collect and compost the food waste. Noise can be controlled by limiting loudspeaker volume and using traditional instruments. Barricades, marked entry and exit lanes, volunteers, and first-aid points can manage the crowd safely.

With these steps, the festival can keep its sacred joy and remain clean and peaceful.

Question 3: Regional folk art and handicraft fairs with renewable energy and zero-waste practices

Folk Art and Handicraft Fairs: Heritage with a Green Future

A regional folk art and handicraft fair is a colourful meeting of tradition and community. Walking through the fair grounds, visitors see Karagattam and Oyilattam performances, hear the beat of the parai drum, and watch artisans make Thanjavur dolls, palm-leaf baskets, clay pots and silk sarees. The air smells of fresh filter coffee, murukku, adhirasam and hot sundal.

These fairs are important because they give artisans a market and help young people see the value of their heritage. Many stalls offer live demonstrations, so children can try pottery or weaving themselves. Local delicacies prepared by village cooks add to the festive atmosphere and support small businesses.

However, large fairs use a lot of electricity and produce a lot of waste. Modern environmental strategies can solve this. Solar micro-grids, made of rooftop panels and batteries, can power the lights, sound systems and stalls without diesel generators. Zero-waste composting can turn leftover food and leaves into manure for local farms. Organisers can use reusable steel plates, clay cups and cloth banners, provide separate bins for wet and dry waste, and recycle paper and cardboard.

If we combine the traditional skill of our artisans with clean energy and responsible waste management, these fairs can protect our culture and our planet together.

Worksheet 2 – Topic 3: Free Writing – Food and Technical Travel Plans

Question 1: Technical study tour to a major hydroelectric dam

A Technical Study Tour to the Mettur Hydroelectric Dam

Our department has planned a one-day technical study tour to the Mettur Dam on the Cauvery River in Salem district. The group of forty students and two faculty members will leave the college by a chartered bus at 4:30 a.m. and travel along the national highway, with a short breakfast halt on the way. The journey takes about six hours, and we will return by night. Students will carry ID cards, water bottles and a notebook.

After registering at the gate, an engineer will take us through the powerhouse. The long concrete dam holds back the huge Stanley Reservoir. Inside the powerhouse, we will see the penstocks, the turbines and the generators, and learn how falling water turns turbines to produce electricity. We will also observe the control room, where operators monitor water level, flow and power output on screens, and the spillway gates used to release extra water safely.

For lunch, we will try local dishes such as hot rice with sambar, fish fry from the river and fresh mango juice, which make the trip a cultural experience as well. In the evening, we will walk through the nearby park and enjoy the green landscape and the calm reservoir.

The visit will help us understand renewable energy, water management and the role of civil, mechanical and electrical engineering in one project. We will follow all safety rules, wear helmets where needed, and stay with our group at all times.

Question 2: Technical study tour to an integrated solar and wind hybrid energy farm

A Technical Study Tour to a Solar and Wind Hybrid Energy Farm

We plan a two-day study tour to the Muppandal wind farm region near Kanyakumari, where wind turbines and solar panels work together. Thirty students and two faculty members will take the overnight train from Chennai to Nagercoil, then travel about one hour by a hired van to the farm. We will stay one night in a budget lodge nearby, and return by train on the second night.

On arrival, a site engineer will give a short briefing on how the hybrid plant works. The wind turbines turn on tall towers, and their blades rotate a generator inside the nacelle. We will watch how the strong coastal wind drives them, and how the blades adjust their angle with the wind speed. In the solar section, we will observe rows of photovoltaic panels, inverters and transformers that feed power into the grid. The control room will show how the two sources balance each other: solar power peaks in the afternoon, whereas wind is strong in the evening and

night.

For food, we will taste Nanjil Nadu specialities such as fish curry with rice, banana chips, avial and fresh coconut sweets. Eating in small local hotels along the route will also help us understand the region's culture.

The tour will show us the practical side of renewable energy, and we will follow safety rules by wearing helmets, keeping away from the turbine bases and obeying the engineers' instructions.

Question 3: Technical field visit to an automated seawater reverse osmosis desalination plant

A Technical Field Visit to a Seawater Reverse Osmosis Desalination Plant

Our class is planning a half-day field visit to a seawater reverse osmosis desalination plant on the East Coast Road near Chennai, such as the Nemmeli plant. About fifty students and three faculty members will travel in two college buses, leaving at 8:00 a.m. and reaching the plant in about ninety minutes. We will carry ID cards and wear closed shoes for safety.

At the plant, the engineers will first explain the process. We will see the intake pumping station, which draws seawater through long pipes under the sea bed. The water then passes through pre-treatment filters that remove sand and suspended particles. Next, high-pressure pumps push the water through the membrane filtration racks, where the salt is separated and fresh water is produced. We will also see the brine discharge outlet, where the concentrated salt water is carefully diluted and released back into the sea to protect marine life. The control room shows the whole process through automated screens.

After the visit, we will have lunch at a local seaside restaurant. The coastal menu includes fish fry, prawn masala, crab curry with steamed rice and filter coffee, which gives us a taste of the Chennai coastal culture.

The visit will help us see how engineering solves water scarcity, how energy is used in the process, and why environmental care in brine disposal is important.

Worksheet 2 – Topic 4: Formal Letters of Request

Question 1: Letter to the Head of the Computer Science Department (lab access for a hackathon)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and lab access after college hours to prepare for a National AI Hackathon

Respected Sir/Madam,

I am writing on behalf of my project team of four first-year students. We have registered for a National AI Hackathon to be held on [date], and we respectfully request permission to use the computer laboratory after college hours.

The objective of our project is to build a simple AI-based solution that solves a real campus problem, using Python and machine learning. We need about three weeks to design, train and test the model, and our home systems are not powerful enough for this.

We require access to the lab between 4:30 p.m. and 7:30 p.m. on weekdays, with Python, Jupyter Notebook, TensorFlow or scikit-learn, Git and a stable internet connection. We promise to follow all lab rules, maintain the equipment carefully, and leave the lab clean and switched off after use. We will also work under the supervision of a faculty member if required.

We would be very grateful if you could grant us this permission.

Thank you for your kind consideration.

Yours obediently,
Inba Ilakiyan
(On behalf of the project team)

Question 2: Letter to the Plant Manager (industrial plant visit)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Plant Manager
[Name of Factory], Industrial Robotics and Automation Division
[Address]

Subject: Request for permission to conduct an industrial visit for 50 first-year engineering students

Respected Sir/Madam,

I am a first-year engineering student of St. Joseph's College of Engineering, Chennai. On behalf of my class, I request your permission to visit your robotics and automation factory with fifty students and two faculty members on a convenient date in [month].

The intended learning outcomes of the visit are to understand how robots and automated systems work on a real production line, to see how sensors, controllers and robotic arms are combined, and to learn about industrial safety practices and quality control. The visit will also help us connect our classroom learning with practical engineering and inspire us to think about careers in automation.

We will arrive in a group, carry our ID cards, follow all safety rules, and obey your staff's instructions throughout. We would be glad if a short briefing and a guided tour of about two hours could be arranged.

I would be grateful if you could kindly grant us this opportunity. I look forward to your reply.

Yours faithfully,
Inba Ilakiyan

Question 3: Letter to the Head of the Mechanical Engineering Department (machine shop access for an electric go-kart)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for permission and machine shop access after college hours to build an electric go-kart

Respected Sir/Madam,

Our student team has been selected to take part in an inter-collegiate design challenge to be held on [date], where we must build and test an energy-efficient electric go-kart. We therefore request permission to use the machine shop after regular college hours.

We need the workshop to cut and weld the chassis, fit the motor, battery and braking system, and test the kart on the campus ground. We estimate that we will need about four weeks, working from 4:30 p.m. to 7:30 p.m. on weekdays and during part of the weekends.

We will work only under the supervision of the workshop instructor, wear safety glasses, gloves and shoes at all times, and follow all machine safety rules. We will take care of the tools and keep the shop clean after use.

We would be thankful if you could grant us this permission and allow us to bring honour to the college.

Thank you for your kind support.

Yours obediently,
Inba Ilakiyan
(On behalf of the team)

Question 4: Letter to the Principal / Dean of Student Affairs (sponsorship and duty leave)

Inba Ilakiyan
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal / Dean of Student Affairs
St. Joseph's College of Engineering
OMR, Chennai – 600 119

Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo

Respected Sir/Madam,

I am happy to inform you that a five-member multidisciplinary team of our college has been selected to present a drone surveillance prototype at the National Defense Innovation Expo, to be held on [dates] at [venue]. We respectfully request financial sponsorship and duty leave for this event.

Our prototype uses a camera-equipped drone with image processing to monitor sensitive areas in real time. Presenting it at a national level will give us valuable feedback from experts and will bring recognition to our college.

We request sponsorship for travel, accommodation, registration and prototype transport, which we estimate at ₹[amount], and duty leave for [number] days so that we do not lose attendance. We will make up for the missed classes and submit a report after the event.

I would be grateful for your kind support and approval.

Thank you.

Yours obediently,
Inba Ilakiyan
(On behalf of the team)

Worksheet 2 – Topic 5: Compare and Contrast Descriptive Essays

Question 1: Electric Vehicles versus Internal Combustion Engine Vehicles

Electric Vehicles versus Conventional Internal Combustion Engine Vehicles

Electric vehicles (EVs) and internal combustion engine (ICE) vehicles are two different answers to the same need, which is transport. Each has clear positives and negatives.

Consider battery life first. An EV battery usually lasts eight to ten years, but replacing it is very expensive. In contrast, an ICE vehicle has no battery problem, though its engine needs regular servicing. As for the manufacturing footprint, EVs create more emissions during production because of battery mining and processing, whereas ICE vehicles have a lighter manufacturing footprint. However, over its lifetime an EV

produces far lower emissions, especially when the electricity comes from renewable sources.

Operating costs also differ. EVs have fewer moving parts, so they need less maintenance, and electricity per kilometre is cheaper than petrol or diesel. On the other hand, ICE vehicles cost less to buy, and their repair shops are available everywhere. Fuel prices, however, keep rising and make them costly to run.

Charging network feasibility is the biggest challenge for EVs. Charging stations are still limited, especially on highways and in small towns, and charging takes much longer than refuelling. ICE vehicles, conversely, can be refuelled in minutes at petrol stations spread across the country.

In conclusion, EVs are cleaner and cheaper to run, whereas ICE vehicles are cheaper to buy and more convenient at present. As battery prices fall and charging networks grow, EVs are likely to become the better choice for the future.

Question 2: Remote/Virtual Internships versus On-Site Internships

Remote Internships versus On-Site Internships for First-Year Engineering Students

Internships help early-stage engineering students connect classroom theory with real work. Today, students can choose between remote (virtual) and on-site (physical) internships, and each has advantages and disadvantages.

Hands-on skill acquisition is the biggest difference. On-site internships give direct access to machines, laboratories and real equipment, so students learn practical skills such as wiring, fabrication and testing. Remote internships, in contrast, mostly involve software, simulations and online tasks, which are useful for coding and design but cannot replace physical practice.

In terms of schedule flexibility, remote internships win clearly. Students can work from home, save travel time and cost, and manage studies alongside the internship. On-site internships, on the other hand, demand fixed hours and daily travel, which can be tiring during the semester.

Mentorship is stronger on-site, because supervisors can guide students directly and correct mistakes immediately. Online mentors, however, are often busy and communicate only through scheduled calls or messages, so doubts may stay unresolved for days.

Professional networking also favours on-site internships, as students meet colleagues, build relationships and learn workplace culture naturally. Remote interns may never meet their team face to face, although online platforms still allow them to connect with professionals from different cities.

In conclusion, remote internships are better for flexibility and software skills, whereas on-site internships are better for practical learning, mentorship and networking. A first-year student should ideally begin with a remote internship and move to an on-site one later.

Question 3: Cloud Computing Infrastructure versus On-Premise Data Centers

Cloud Computing versus On-Premise Physical Data Centers for Engineering Enterprises

Engineering enterprises must decide whether to run their systems on the cloud or in their own physical data centers. Both choices have strengths and weaknesses.

Upfront capital expenditure is the clearest difference. Cloud services need almost no initial investment, because companies simply pay for what they use. On-premise data centers, in contrast, require heavy spending on servers, cooling systems, power backup and buildings before any work begins.

As for recurring overheads, cloud bills can grow quickly if usage is not monitored, whereas an on-premise setup has predictable costs once it is built, although it still needs staff, electricity and maintenance. Therefore, cloud suits changing workloads, while on-premise suits stable and long-term demand.

Workload scalability strongly favours the cloud. Companies can add computing power in minutes during simulations or product launches and reduce it afterwards. On the other hand, on-premise systems have fixed capacity, so expanding them takes weeks or months and leaves hardware idle during quiet periods.

Data security and compliance are more complex. On-premise data centers give the company full physical control, which is important for defence, healthcare or confidential design data. Cloud providers offer strong security tools and certifications, but the company depends on a third party and must follow data-location laws

carefully.

In conclusion, the cloud is better for flexibility, speed and low starting cost, while on-premise infrastructure is better for control and predictable long-term expenses. Many enterprises now choose a hybrid model that combines the advantages of both.

Question 4: 3D Printing versus Traditional CNC Milling

3D Printing (Additive Manufacturing) versus CNC Milling (Subtractive Manufacturing) in Prototype Fabrication

In prototype fabrication, 3D printing and CNC milling follow opposite principles. A 3D printer builds a part layer by layer, whereas a CNC machine cuts material away from a solid block. Each method has positives and negatives.

Material wastage is the first point of contrast. 3D printing uses only the material needed for the part, so wastage is low. CNC milling, on the other hand, removes a large amount of material as chips, which increases the cost of expensive metals.

When it comes to production speed, the picture changes with batch size. For single prototypes and small batches, 3D printing is faster because it needs no tooling or setup. For larger batch runs, however, CNC milling is quicker, as it can produce identical parts continuously at high speed.

Surface finish and accuracy favour CNC milling. It produces smooth surfaces and very tight tolerances, so parts often need no extra work. In contrast, 3D printed parts often show layer lines and need sanding or polishing, and their accuracy is lower.

Geometric complexity is the strength of 3D printing. It can easily create hollow parts, internal channels and lattice structures that are impossible to cut with a rotating tool. CNC milling is limited by tool access and the direction of cutting.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with little waste, whereas CNC milling is better for accurate, smooth and larger-batch parts. Engineers often use both methods together to get the best result.

Worksheet 2 – Topic 6: Personal Email Writing

Question 1: Email to a high school mentor/teacher about the first semester

To: [teacher's email address]

Subject: My first semester at engineering college

Dear Sir/Madam,

I hope you are doing well. I wanted to write to you because your guidance in school gave me the confidence to start my engineering journey, and I would like to share how my first semester has been.

The biggest change was the move to independent campus life. Nobody reminds us about deadlines now, so I had to learn to manage my own time, plan my study hours and take responsibility for my work. It was difficult at first, but I am getting used to it.

In our laboratory sessions, we worked on our first small projects, such as basic programs in Python and C and simple circuits. Seeing my own code run correctly gave me a great sense of satisfaction, and I realised how different practical learning is from reading theory.

For the end-semester examinations, I have made a timetable that gives equal time to every subject. I revise the notes every weekend, solve previous question papers and practise the lab experiments so that I do not forget the procedures.

Thank you for everything you taught me. I will be very happy to hear from you and to receive any advice for the coming semesters.

With respect and regards,
Inba Ilakiyan

Question 2: Email to a high school science teacher about the first term

To: [science teacher's email address]

Subject: Reflections on my first term in engineering

Dear Sir/Madam,

I hope you are keeping well. It has been a few months since I joined engineering college, and I wanted to share my first-term experience with you, since you were the one who made science exciting for me.

The first thing I noticed was how different the hands-on lab coursework is. In school, we mostly saw experiments in a few periods, but here we handle components, tools and software ourselves. I made mistakes in the beginning, but each mistake taught me something and I am becoming more careful and confident.

I also took part in my first technical club activity and a small hackathon. Working with a team to build a solution within a limited time was challenging and thrilling, and I learned a lot about teamwork and debugging from my seniors.

For the upcoming university theory and practical exams, my strategy is simple. I will finish revising each unit at least two weeks before the exams, practise numerical problems daily, and repeat the lab experiments so that I can answer viva questions without fear.

I would be grateful for any advice you may have. Please take care.

Yours sincerely,
Inba Ilakiyan

Question 3: Email to an elder sibling/alumnus seeking advice

To: [sibling's or alumnus's email address]

Subject: Need your advice on balancing studies and technical clubs

Dear [Name],

How are you? I hope college and work are going well. I am writing because I need your advice on something I am struggling with.

My day starts at 6:30 a.m., and I attend classes until 4:30 p.m. After that, I go to my technical club meetings for about two hours, and in the evening I try to study and finish my assignments. By the time I am free, I am usually very tired.

My biggest challenge is managing the lab record submissions. Because the records must be completed every week, they pile up, and I often finish them at the last minute, which affects the quality.

Since you have handled the same situation, could you please tell me how you balanced your academics with the clubs? I would also like your tips for semester exam revision, such as how to plan the timetable, which subjects to start with, and how to use previous question papers.

I am looking forward to your reply. Thank you in advance, and please take care.

Your loving brother/sister,
Inba

Worksheet 2 – Topic 7: Travel Planning and Budgetary Itineraries

Question 1: 4-day tour to an aerospace and satellite launch station

A 4-Day Student Technical Tour to an Aerospace and Satellite Launch Station

Our department plans a four-day tour to a satellite launch station for forty students and two faculty members. We will travel by a chartered train coach and local bus, so that the cost is reasonable.

Budget (per student): train fare ₹1,200, local bus ₹400, accommodation for three nights ₹1,500, food ₹1,200, entry and miscellaneous charges ₹400. The total is ₹4,700, which gives about ₹1,88,000 for the full group of forty.

Day 1: Overnight travel and arrival, check-in at the guest house and an orientation briefing. **Day 2:** Visit the launch complex viewing gallery, the rocket assembly area and the mission control centre, where scientists explain how a launch is planned and tracked. **Day 3:** Visit the satellite integration and testing laboratory and the space museum, followed by an interaction with engineers on careers in aerospace. **Day 4:** Morning group discussion, preparation of our tour report, and the return journey.

Safety protocols: Carry your ID cards and keep them visible at all times. Follow the instructions of the security staff, avoid photography in restricted areas, and stay with the group. Wear closed shoes and drink plenty of water. Keep the emergency contact numbers of the faculty coordinators and follow the launch-site rules strictly.

This tour will give us a first-hand view of how rockets and satellites are built and operated.

Question 2: 4-day tour to a major commercial port and maritime dry-dock facility

A 4-Day Industrial Exposure Tour to a Major Commercial Port and Dry-Dock Facility

Our class of forty first-year students and two faculty members will visit a major commercial port and a dry-dock facility over four days. We will travel in a hired bus, and stay in a budget hotel near the port.

Estimated cost per student: travel ₹1,500, stay for three nights ₹1,800, food ₹1,200, entry passes and miscellaneous ₹500. The total is ₹5,000 per student.

Day 1: Travel to the port city, check in and a short orientation. **Day 2:** Technical briefing at the port authority, followed by a visit to the container terminal, where cranes, conveyors and tracking systems handle cargo. **Day 3:** Visit the dry-dock, where ships are lifted and repaired, and see how hull cleaning, welding and engine maintenance are done. An engineer will also brief us on ship stability. **Day 4:** Visit the control tower and the vessel traffic system, a feedback session and the return journey.

Dockyard safety protocols: Wear helmets, safety shoes and high-visibility jackets at all times. Do not go near moving cranes, cargo or the edge of the dock. Stay in the group, obey the guides and never touch machinery. Keep emergency contacts ready, and carry first-aid supplies.

This tour will help us understand how ports and shipyards work in real life.

Question 3: 4-day tour to an electric locomotive manufacturing works and railway signalling center

A 4-Day Technical Study Tour to an Electric Locomotive Works and a Railway Signalling Center

We plan a four-day study tour for thirty-five students and two faculty members to an electric locomotive manufacturing works and an automated railway signalling center. We will book sleeper-class seats on trains for both journeys, and use a local bus for the visits.

Category-wise expenditure per student: rail tickets ₹1,400, local transport ₹300, accommodation for three nights ₹1,500, food ₹1,200, and entry and miscellaneous ₹300. The total is ₹4,700.

Day 1: Overnight train journey and arrival. **Day 2:** Visit the locomotive works, where we will see the shop-floor stages such as shell fabrication, wiring, traction motor assembly and final testing of the engine. **Day 3:** Visit the automated signalling center, where engineers explain how sensors, track circuits and computers control train movement and keep the trains safe. **Day 4:** Interaction with engineers, preparation of a tour report and the return journey.

Site safety rules: Wear helmets, safety shoes and closed clothing in the shop floor areas. Never walk on the railway tracks, and keep a safe distance from cranes and live electrical equipment. Follow the guides' directions and stay with the group. Do not touch any machine or signal equipment, and report any problem immediately to the faculty coordinator.

This tour will connect our classroom learning with real railway engineering.

Vocabulary Worksheet – Solved

Business vocabulary: Units 1 to 63 (answers underlined)

Part A: Fill in the Blanks

Complete the sentences using the correct terms from Units 1, 2, and 3.

To find out what someone's job is, you usually ask, “What **do** you **do**?”

A manager who oversees a team and ensures targets are met is often said to be **in charge** of the department.

Employees who work outside standard office hours in rotation (e.g., one week morning, next week evening) are working in **shifts**.

Outside specialists hired by companies to persuade top executives to leave their current firms are known as **headhunters**.

Job applicants typically submit a **CV** (curriculum vitae) detailing the story of their working life alongside a covering letter.

Part B: Matching Terms to Definitions

Match each business term on the left with its correct description on the right.

B Flexitime

C Teleworking

E Psychometric test

D Shortlist

A Referees

A. Previous employers or teachers named to provide character or performance references.

B. A system allowing employees to choose working hours within agreed limits.

C. Working from home using a computer and telecommunications.

D. A selected pool of qualified candidates chosen for final interviews.

E. A written test used to assess a candidate's intelligence and personality.

Part C: Sentence Correction

Identify and correct the errors in the following sentences.

Error: "I am in charge for the design department." Correction: **I am in charge of the design department.**

Error: "He is going to the work by train every morning." Correction: **He is going to work by train every morning.**

Error: "She is a very responsible in her daily tasks." Correction: **She is very responsible in her daily tasks.**

Part A: Fill in the Blanks

Complete the sentences using the correct terms from Units 4 and 5.

Someone who has just left university with a degree is referred to as a **graduate**.

A worker who requires virtually no specialized training or formal qualifications is classified as **unskilled**.

Salespeople often receive a basic salary plus **commission**, which is a percentage on everything they sell.

Senior executives may receive **stock** options, giving them the right to buy company shares at low prices.

High-ranking corporate executives with excessive compensation packages are sometimes referred to informally as **fat cats**.

Part B: Matching Collocations

Match the adjectives/phrases on the left with their correct prepositional complements.

D Skilled **at / in**

E Good **with**

A Computer **literate**

B Self **starter**

C Fringe **benefits**

A. literate

B. starter / motivated

C. benefits

D. at / in

E. with

Part C: Application & Context Analysis

Explain the difference between the following pairs in 1-2 sentences:

Salary vs. Wages: **Salary is usually a fixed amount paid regularly, often monthly or annually. Wages are generally based on hours worked or the amount of work completed.**

Severance package vs. Bonus: **A severance package is compensation and benefits given when employment ends. A bonus is extra pay awarded in addition to normal salary or wages, often for performance.**

Part A: Fill in the Blanks

Complete the sentences using terms from Units 6, 7, and 8.

Manual workers who use their hands and wear blue collars are often contrasted with **white-collar** workers in offices.

When workers stop working temporarily to protest pay or conditions, they go on **strike**.

A company that reduces its management layers and makes itself leaner is said to have **downsized** and **delayered**.

When an employee is dismissed through no fault of their own because their job no longer exists, they are made **redundant**.

An invisible barrier that prevents women from rising beyond a certain corporate level is called the **glass** ceiling.

Part B: Matching Health & Safety / Legal Terms

Match each workplace issue with its accurate classification.

B Repetitive Strain Injury (RSI)

D Passive smoking

A Sexual harassment

C Equal opportunities

E Whistleblowing

A. Unwelcome sexual behavior in the workplace

B. Physical injury caused by continuous data entry or typing

C. British term equivalent to US affirmative action

D. Breathing in other people's tobacco smoke involuntarily

E. Reporting illegal or unethical practices within an organization

Part C: Short Answer Exercises

What is the difference between being dismissed and being made redundant? **Being dismissed means an employee is removed from a job, usually for a reason connected with conduct or performance. Being made redundant means the job itself is no longer needed, so the employee loses the job through no fault of their own.**

What is an open-plan office? **An open-plan office is a workplace with a large shared area rather than separate offices or rooms for each employee.**

Part A: Fill in the Blanks

Complete the sentences using terms from Units 9, 10, and 11.

Directors who are outsiders and do not participate in the day-to-day running of the company are **non-executive** directors.

In the US, the top manager is often the CEO, while **executives** take care of the day-to-day running.

Someone who founds or establishes their own company and takes financial risks is an **entrepreneur**.

Powerful leaders who control massive business empires are often called magnates, moguls, or **tycoons**.

Businesses that are sold off by the state to private investors are **privatized**.

Part B: Matching Corporate Terms

Match the terms on the left with their definitions.

A SME

D Corporate culture

E Enterprise zone
B Head up
C Boardroom
A. Small and Medium-sized Enterprise
B. To head or lead a department or activity
C. The room where directors meet
D. The shared beliefs and working style of a company
E. A designated region with fewer laws and lower taxes to encourage business

Part C: Error Correction

Correct the italicized phrases:
"He is the head up of our marketing department." **He heads up our marketing department.**
"We need to buy a commerce for our new online shop." **We need to buy an e-commerce platform for our new online shop.**

Part A: Fill in the Blanks

Complete the sentences using terms from Units 12, 13, and 14.
A self-employed person working alone, especially a builder or plumber, is **a sole trader**.
A company whose shares can be freely bought and sold on the stock market is a Public Limited Company or **PLC**.
Organizations with social aims like helping the sick or poor are non-profit organizations or **charities**.
Software developers often release a final test version, called a **beta** version, before official launch.
If a design defect is discovered after a product launch, a company may have to **recall** it.

Part B: Sector Classification

Classify each industry as either Manufacturing (M) or Services (S).
M Aerospace
S Catering
M Pharmaceuticals
S Property / Real Estate
M Steel production

Part C: Short Definitions

Define CADCAM: **Computer-Aided Design and Computer-Aided Manufacturing: using computers to design products and control manufacturing processes.**
Define Focus group: **A small group of people brought together to discuss and give opinions about a product, service, or idea.**

Part A: Fill in the Blanks

Complete the sentences using terms from Units 15, 16, and 17.
Products that are no longer up-to-date because they use old technology are **obsolete**.
Legal protection for an invention that stops others from copying its design is a **patent**.
The maximum amount that a plant or company can produce is its **capacity**.
Materials, components, labor, and capital used in production are collectively known as **inputs**.
Manufacturing goods efficiently without waste by having components arrive exactly when needed is **just-in-time** production.

Part B: Matching Terms

B R&D
A Royalties
C Craft industry
E Work-in-progress
D Outsourcing
A. Payments made to an author or inventor for using their protected work

B. Research and Development department
C. Manual, labor-intensive production of items like furniture
D. Using outside suppliers for services previously done in-house
E. Goods that are currently being made on the production line

Part C: Short Explanations

Why do companies implement Just-in-Time (JIT) manufacturing? **Companies implement JIT to reduce inventory and waste, lower storage costs, and receive components when they are needed for production.**

Part A: Fill in the Blanks

Complete the sentences using terms from Units 18, 19, and 20.
The objective of quality control is **conformance** to specifications with zero defects.
Comparing your performance against the best operators in the industry is called **benchmarking**.
People who buy everyday services like train travel are customers, while those who buy professional services are **clients**.
A company that sells more than any other competitor in a market is the market **leader**.
When competition is extremely strong and aggressive, it is often described as **cut-throat** or fierce.

Part B: Matching Terms

B Kaizen
C Mystery shopper
E Market segment
D Key players
A BPR
A. Business Process Re-engineering
B. Japanese term for continuous improvement
C. Person pretending to be a customer to check service quality
D. The most important companies in a particular market
E. A distinct group of buyers targeted within a market

Part C: Analysis Task

What is the difference between a market leader and a challenger? **A market leader has the largest market share. A challenger is a competitor trying to increase its market share and challenge the leader.**

Part A: Fill in the Blanks

The four Ps of marketing are product, price, **place**, and promotion.
Consumer goods that last a long time, such as cars and washing machines, are consumer **durables**.
Products that do not have a brand name are known as **unbranded** products or generics.
Cheap items sold at or below cost to attract customers into a store are **loss** leaders.
A small group of buyers with specialized needs forms a **niche** market.

Part B: Matching Terms

A FMCG
B Own-brand
D Upmarket
C Trade up
E Price war
A. Fast-Moving Consumer Goods (fast-selling items like food)
B. Products sold under the retailer's own name
C. Moving from basic models to more sophisticated, expensive ones
D. Designed for higher-income users; upscale
E. When competing companies continually reduce prices in response to each other

Part C: Short Answer

Name the Four Ps of the marketing mix: **Product, Price, Place and Promotion.**

Part A: Fill in the Blanks

Complete the sentences using terms from Units 24, 25, and 26.

Wholesalers and retailers are collectively known as **distributors**.

A retail store that is part of a group of shops all sharing the same name is a **chain** store.

Unsolicited direct mail sent to consumers' homes is often dismissively called **junk** mail.

A series of advertisements for a particular product or company is an advertising **campaign**.

Businesses combining traditional physical shops with online e-commerce operations practice **clicks**-and-mortar retail.

Part B: Matching Terms

C Cold call

E Franchise

A Billboard (AmE)

D Loyalty card

B B2B

A. Open-air hoarding used for advertising

B. Business-to-Business e-procurement

C. Unannounced sales phone call to someone with no prior contact

D. Card offering reward points based on customer spending

E. Business format run by a franchisee under a franchisor's brand

Part A: Fill in the Blanks

Money received from sales is known as revenue or **turnover**.

Costs that do not change when production goes up or down (e.g., rent) are **fixed** costs.

Products that have very good profitability and generate large cash surpluses are cash **cows**.

Customers who owe money to a business are recorded as accounts **receivable**.

Independent outside accountants who examine a company's accounts to see if they give a true and fair view are **auditors**.

Part B: Matching Financial Terms

A COGS

C Goodwill

B Depreciation

E In the red

D Window dressing

A. Cost of Goods Sold (variable manufacturing costs)

B. Reduction in the book value of fixed assets over time

C. An intangible asset representing a company's good reputation

D. Creative accounting making results look better than they are

E. Operating at a financial loss or having a negative bank balance

Part A: Fill in the Blanks

1. The person who is in charge of running a meeting and keeping participants on track is the chairperson or chair, often referred to simply as the **chair**.

2. A list of items to be discussed during a meeting, sent out in advance, is called the **agenda**.

3. When negotiators or meeting participants reach a middle-ground solution where both sides make concessions, they reach a **compromise**.

4. Using cautious language to avoid making overly direct or absolute statements is known in discussion techniques as **hedging**.

5. When a meeting discussion completely breaks down and no agreement can be reached, it is referred to as a **deadlock**.

Part B: Matching Meeting Terms

1. **C** AGM
2. **E** Minutes
3. **A** Consensus
4. **B** Action points
5. **D** Hedging

· A. General agreement reached by all participants
· B. Specific tasks assigned to individuals to complete after a meeting finishes
· C. Annual General Meeting held once a year for shareholders
· D. Avoiding direct or committal statements using qualifying language
· E. The written record of decisions and discussions made during a meeting

Part C: Short Answer Exercises

1. What is the main responsibility of a meeting chairperson before a meeting begins? **To prepare and organize the meeting, including setting the agenda and ensuring participants know what will be discussed.**

2. What is the difference between reaching a consensus and voting on a decision? **Consensus means the participants reach general agreement together. Voting makes a decision according to the choice receiving the required majority.**

Part A: Fill in the Blanks

1. The visual items used during a presentation, such as slides, handouts, or overhead projections, are called visual **aids**.
2. Establishing a positive connection and good mutual relationship with your audience during a presentation is building **rapport**.
3. The opening part of a presentation where you welcome the audience and outline your structure is the **introduction**.
4. Moving around too much or making nervous hand gestures can distract from your body **language** during a speech.
5. When concluding a presentation, speakers typically open the floor to deal with questions from the **audience**.

Part B: Matching Presentation Stages

1. **C** Visual aids
2. **B** Body language
3. **D** Rapport
4. **E** Q&A session
5. **A** Handouts

· A. Printed materials distributed to listeners to supplement a presentation
· B. Postures, gestures, and facial expressions used while speaking
· C. Charts, slides, and diagrams used to illustrate key points visually
· D. A harmonious, trusting relationship between speaker and listeners
· E. The time at the end of a presentation dedicated to answering inquiries

Part C: Error Correction

Correct the italicized phrases:

1. "During my presentation, I will show you some visual supports." **During my presentation, I will show you some visual aids.**

2. "Let's open the floor for some inquiries from the spectators." **Let’s open the floor for some questions from the audience.**

Part A: Fill in the Blanks

Complete the sentences using terms from Units 62 through 66.

1. A negotiation style where both sides aim for an outcome that benefits everyone involved is known as a **win** -win situation.
2. Offering something you value less in exchange for something the other party values less during talks is making **trade** -offs.
3. When negotiations completely stall and neither side is willing to move or compromise, they have reached a **deadlock** .
4. An agreement between two or more parties that is enforceable by law after a successful negotiation is a **contract** .
5. A neutral third party brought in to help resolve a deadlock and mediate an agreement is a **mediator** .

Part B: Matching Negotiation Tactics

1. **A** Probing
2. **D** Counter-proposal
3. **B** Deadlock
4. **E** Concession
5. **C** Mediator

· A. Asking probing questions to discover the other party's underlying needs
· B. A stall in talks where neither side will compromise
· C. An independent third party assisting in dispute resolution
· DT. A responding offer made in reaction to an initial proposal
· E. Something given up voluntarily to reach an agreement

Part C: Short Answer Exercises

1. What is the difference between a hard bargaining style and a win-win style? **Hard bargaining focuses on gaining as much as possible, often through firm or competitive tactics. A win-win style seeks solutions that provide acceptable benefits to both sides.**
2. Why is checking the final contract details essential after reaching an agreement? **It ensures that all agreed terms are accurately recorded, understood, and legally clear, reducing the risk of disputes or misunderstandings later.**

Part A: Fill in the Blanks

Complete the sentences using terms from Units 36 and 37.

1. When a company sells its shares to the public for the first time, it floats the company in a **flotation** .
2. Short-term lending to businesses is known as commercial **credit** .
3. Financial products derived from underlying assets, such as futures and options contracts, are known as **derivatives** .
4. An extended period of rising share prices and optimistic dealer sentiment is referred to as a **bull** market.
5. Shares in long-established, stable companies with a history of profit in both good and bad times are called **blue** chips.

Part B: Matching Financial Markets

1. **B** Forex
2. **C** Bourse
3. **A** NASDAQ
4. **E** FTSE
5. **D** Bear market

· A. New York stock market index focusing heavily on high-tech companies
· B. Foreign exchange market for buying and selling national currencies
· C. An alternative term for a stock market or stock exchange
· D. A market characterized by falling prices and pessimistic dealer sentiment

· E. London's Financial Times Stock Exchange index

Part C: Short Answer Exercises

1. What is the difference between a futures contract and an options contract? **A futures contract creates an obligation for the parties to buy or sell the underlying asset at an agreed price and date. An option gives the holder the right, but not the obligation, to buy or sell.**

2. Why are major financial centres like London and New York often referred to by nicknames like "The City" or "Wall Street"? **These nicknames refer to important financial districts or centres that are strongly associated with banking, investment, stock exchanges, and financial services.**

Part A: Fill in the Blanks

1. The study of how money works, is used, and whether economic activities are profitable is called **economics**.
2. When a country exports more goods and services than it imports, it maintains a trade **surplus**.
3. The total monetary value of all goods and services produced within a particular country is its Gross **Domestic** Product (GDP).
4. A period of negative economic growth where economic output contracts is known as a **recession**.
5. When a bad recession becomes extremely severe, widespread, and prolonged, it is called a **slump** or depression.

Part B: Matching Economic Indicators

1. **C** Inflation rate
2. **D** Trade deficit
3. **E** Stagflation
4. **B** Trough
5. **A** Stagnation

· A. A period when the economy is growing very slowly or not at all
· B. The lowest point in an economic cycle before output starts rising again
· C. The speed at which general consumer price levels are rising
· D. When import payments exceed export earnings, creating a trade gap
· E. A difficult economic condition combining slow growth with rapid inflation

Part C: Error Correction

Correct the italicized phrases:

1. "The country's economic indicators show a high rate of inflation speeds." **The country's economic indicators show a high rate of inflation.**
2. "When output reaches its lowest level, it hits a peak before rising." **When output reaches its lowest level, it hits a trough before rising.**

Part A: Fill in the Blanks

1. Buying or selling securities using confidential information that is not publicly available is illegal **insider** dealing.
2. A secret agreement among competing companies in the same market to fix prices at a certain level is price **fixing**.
3. A type of fraud where an employee illegally steals and gets money from their employer is known as **embezzlement**.
4. The process of hiding the illegal origin of criminal money, often by purchasing property, is money **laundering**.
5. Directing investment funds exclusively into companies that do not harm people or the environment is ethical **investment**.

Part B: Matching Crime & Ethics Terms

1. **D** Embezzlement
2. **A** Counterfeiting
3. **B** Bribery

4. **E** Code of ethics
5. **C** Sweatshop labor

· A. Making illegal fake copies of banknotes, documents, or luxury goods
· B. Offering or receiving illegal payments to persuade someone to act corruptly
· C. Manual labor performed under severely exploitative, underpaid conditions
· D. The illegal theft of funds by someone entrusted with managing them within a firm
· E. A formal corporate document outlining required moral standards of behavior

Part C: Short Answer Exercises

1. What is the purpose of establishing Chinese walls within financial institutions? **To prevent confidential information from passing between departments where a conflict of interest could arise, especially between teams handling sensitive information.**
2. Why do institutional investors conduct annual social performance audits? **To assess how companies perform on social and ethical issues and to check whether their practices meet expected standards.**

Part A: Fill in the Blanks

1. The overall period during which a project or task should be completed is referred to as the timescale or **time frame**.
2. When work is completed later than the planned completion date, it is behind schedule and there is a **delay**.
3. Work that is difficult in an interesting, rewarding, and enjoyable way is described as **challenging**.
4. An extreme state of physical and mental exhaustion caused by chronic work stress is known as **burnout**.
5. The metaphor used to describe the exhausting, highly competitive modern corporate work environment is the rat **race**.

Part B: Matching Time & Stress Terms

1. **D** Downtime
2. **E** Prioritize
3. **C** Perfectionism
4. **A** Downshifting
5. **B** Stress-induced

· A. Choosing to leave behind a stressful career for a simpler, balanced lifestyle
· B. An unexpected illness or condition brought on by excessive pressure
· C. An unworkable aim for flawlessness when there is no need for it
· D. Periods when machinery or computers cannot operate because they are broken
· E. Arranging tasks in order of importance before working on them

Part C: Short Answer Exercises

1. What are the key dangers of striving for perfectionism when managing daily tasks? **Perfectionism can waste time, increase stress, delay completion, and make it difficult to prioritize important tasks.**
2. How do downshifters typically redefine their personal idea of success? **They generally place greater value on balance, health, personal time, relationships, and a simpler lifestyle rather than only career status or income.**

Part A: Fill in the Blanks

1. People working below a manager in an organizational hierarchy are known as **subordinates**.
2. A leadership quality that is forward-thinking, with the power to see clearly how things will be in the future, is called being a **visionary**.
3. Management decisions that are arrived at by asking employees to contribute in a process of consultation are built on **consensus**.
4. Granting employees the initiative and authority to make decisions without asking managers first is called **empowerment**.
5. A corporate environment characterized by male dominance, aggressiveness, and physical strength is a **macho** culture.

Part B: Matching Leadership & Cultural Terms

1. **B** Authoritarian
2. **E** Decentralized
3. **D** Deference
4. **C** Corporate culture
5. **A** Canteen culture

· A. Unofficial, unapproved ways that people within an organization think and talk
· B. A management style that imposes decisions from above without consultation
· C. The shared values, habits, and working style of a specific company
· D. Respectful submission and yielding to the authority of superiors
· E. Organizational structures where decision-making authority is spread out

Part C: Error Correction

Correct the italicized phrases:

1. "The manager is very remote, so she is completely approachable by all staff." **The manager is very remote, so she is not approachable by all staff.**
2. "We operate a top-down consensus approach in our department." **We operate a bottom-up, consultative approach in our department.**

Part A: Fill in the Blanks

Complete the sentences using terms from Units 46, 47, and 48.

1. In the English-speaking business world, people routinely use **first** names even when they do not know each other very well.
2. A company day where employees are allowed to wear less formal clothing than a standard business suit is called a **dress-down** Friday.
3. Being at work when you do not actually need to be there, simply to demonstrate commitment, is known as a culture of **presenteeism** .
4. A phone extension that is not connected by a physical wire, allowing movement around a home or office, is a **cordless** phone.
5. A phone number that provides free access for callers, such as an 0800 number, is a **toll** -free number.

Part B: Matching Etiquette Terms

1. **B** Surname
2. **E** Smart casual
3. **D** Punctuality
4. **C** Pager
5. **A** Videoconferencing

· A. Holding a face-to-face meeting across locations using computers and cameras
· B. A family name shared across relatives
· C. A device that receives written messages while on the move
· D. Arriving at meetings or appointments on time
· E. A relaxed yet professional style of dress that avoids full suits

Part C: Short Answer Exercises

1. What is the standard convention regarding names in the English-speaking business world? **People commonly use first names in business interactions, even when they do not know each other very well.**
2. How do attitudes toward time and schedules vary across different business cultures? **Some cultures place strong emphasis on punctuality and strict schedules, while others may take a more flexible approach to time.**

Part A: Fill in the Blanks

1. A formal meeting open to all shareholders of a company held once a year is an Annual General Meeting or **AGM** .

2. Before a meeting begins, the chairperson prepares a list of items to be discussed, known as the **agenda**.

3. During a meeting, the person taking notes records the decisions and discussions to produce the official **minutes**.

4. When a chairperson wants to open a meeting or direct attention to the first item, they formally **open** the meeting.

5. Inviting specific participants to speak during a discussion helps ensure everyone has an opportunity to share their **views**.

Part B: Matching Meeting Terms

1. **C** Chairperson
2. **E** Agenda
3. **D** Minutes
4. **B** Action points
5. **A** Venue

· A. The official physical or virtual location where a meeting takes place
· B. Specific tasks assigned to individuals to execute after the meeting concludes
· C. The person responsible for running the meeting and keeping order
· D. The written record of what was discussed and decided
· E. The structured list of topics scheduled for discussion

Part C: Short Answer Exercises

1. What are the key responsibilities of a chairperson during a meeting? **To keep the meeting organized and on track, manage discussion, give participants opportunities to speak, maintain order, and guide the group toward decisions.**

2. Why is it important to distribute an agenda well in advance of a meeting? **It allows participants to prepare, understand the topics, gather information, and plan their contributions before the meeting.**

Part A: Fill in the Blanks

1. Using cautious language and qualifying phrases to avoid making overly direct or absolute statements is called **hedging**.

2. A negotiation approach where both parties work to find solutions that benefit everyone involved is a **win**-win situation.

3. Asking detailed and probing questions during negotiations helps uncover the other party's underlying **interests**.

4. When negotiators offer something they value less in exchange for a concession from the other side, they are making **trade**-offs.

5. When a negotiation stalls completely because neither side will compromise or move forward, they have reached a **deadlock**.

Part B: Matching Discussion & Negotiation Terms

1. **D** Hedging
2. **C** Consensus
3. **E** Probing
4. **B** Counter-proposal
5. **A** Mediator

· A. An independent third party brought in to help resolve a deadlock
· B. A responding offer made in reaction to an initial bargaining position
· C. General agreement reached collectively by all participants
· D. Using cautious, non-committal language to soften statements
· E. Asking exploratory questions to investigate underlying interests

Part C: Error Correction

Correct the italicized phrases:

1. "We reached a deadlock, so we made a win-win compromise that pleased nobody." **We reached a deadlock, so we could not reach a win-win compromise.**

2. "During the meeting, he used too much hedging so his point was extremely direct and clear." **During the meeting, he used too much hedging, so his point was not direct or clear.**

Part A: Fill in the Blanks

1. A brief presentation given to inform colleagues or clients about a specific project update or status report is often structured around different types of presentations, such as a product **presentation**.

2. When preparing a presentation, speakers should practice to ensure proper timing, avoiding rushing through the material or exceeding the allocated **time**.

3. Maintaining eye contact, varying vocal tone, and projecting your voice properly are all vital elements of managing your **voice** during a speech.

4. Building positive engagement and mutual connection with your audience is essential for establishing good **rapport** early in your presentation.

5. Visual materials such as slides, handouts, and diagrams used to illustrate key concepts during a speech are called visual **aids**.

Part B: Matching Presentation Terms

1. **D** Preparation
2. **A** Rapport
3. **B** Body language
4. **C** Visual aids
5. **E** Q&A session

· A. A harmonious, trusting relationship established between speaker and listeners
· B. Postures, gestures, and physical movements used while speaking
· C. Charts, slides, and diagrams used to clarify information visually
· D. The structured process of planning, drafting, and organizing your speech
· E. The concluding segment dedicated to answering listener inquiries

Part C: Short Answer Exercises

1. What are the key dos and don'ts when preparing the introduction of a presentation? **Do greet the audience, introduce the topic, state the purpose, and briefly outline the structure. Do not overload the introduction with unnecessary detail.**

2. How should a speaker handle difficult or hostile questions during the final question-and-answer period? **Stay calm and professional, listen carefully, acknowledge the concern, answer clearly when possible, and avoid becoming defensive or argumentative.**

Part A: Fill in the Blanks

Complete the sentences using terms from Units 62 and 63.

1. A negotiation where two parties try to reach an agreement over a contract or dispute is heavily influenced by the initial **stage** of negotiation.

2. Before entering formal talks, experienced negotiators spend significant time preparing their strategy, identifying their objectives and acceptable **walk-away** points.

3. The bargaining stage involves making concessions and counter-proposals to move the discussion closer to a mutually acceptable **agreement**.

4. Different negotiators exhibit different negotiating styles, ranging from cooperative and collaborative to highly confrontational and hard **bargaining**.

5. When preparing for a complex international deal, understanding cultural nuances is just as critical as analyzing the financial terms of the **deal**.

Part B: Matching Negotiation Terms

1. **C** Bargaining
2. **D** Concession
3. **A** Counter-proposal

4. **E** Negotiating style
5. **B** Deal

· A. A responding offer made in reaction to an initial bargaining position
· B. An agreement or contract reached after successful discussions
· C. The active discussion process of trading concessions to reach agreement
· D. Something given up voluntarily by one party to encourage agreement
· E. The characteristic approach or behavioral pattern adopted by a negotiator

Part C: Error Correction

Correct the italicized phrases:

1. "During the bargaining phase, we refused to make any concessions and immediately signed the contract." **During the bargaining phase, we made concessions and negotiated before signing the contract.**

2. "His negotiating style was entirely cooperative, meaning he shouted down every proposal made by the other side." **His negotiating style was confrontational, because he shouted down every proposal made by the other side.**

Project Proposal: Ideation

CricScore: Technical Specification | Individual project | Jashwanth S, CSE-C | Draft v1

Status: draft. Items marked [default] are working assumptions that can be changed. Items marked [open] still need a decision.

1. Overview

CricScore is a live match-scoring app for local and college cricket. Scoring such matches on paper is slow and error-prone: runs, wickets and overs are often miscounted, and players and spectators cannot easily see the live score or run rate. CricScore records a match ball by ball, updates runs, wickets, overs and run rate automatically, and produces a final match summary.

The first version is built in Python as a console application and stores match data in CSV files. The design leaves room for a graphical interface, player statistics and a mobile version later.

2. Goals and Non-Goals

2.1 Goals

- Record a limited-overs match ball by ball (runs, wicket, wide, no-ball).
- Update score, wickets, overs and run rate automatically after every ball, and show the required rate in the second innings.
- Show a final scorecard and save every match to a CSV file for easy review.
- Be simple enough that a first-time user can score a match after a short instruction.
- Keep scoring rules, data storage and screen output in separate modules.

2.2 Non-goals (first version)

- No graphical interface, mobile app, accounts or online features.
- No advanced rules such as rain-affected targets or super overs.

3. Target Users and Design Feel

- Users: local and college cricket scorers, with players and spectators following the match.
- Feel [default]: fast and forgiving. Every prompt says what to type, and a wrong entry shows a clear message instead of stopping the program.

4. Technology Stack

Item	Decision
Language	Python 3 (minimum tested version to be recorded in the README)
Interface	Console first; graphical interface is a later stage
Storage	CSV files using Python’s built-in csv module
Testing	pytest (development only)
Version control	Git and GitHub
Platforms	Windows, macOS and Linux, run from source

Run command [default]: python3 main.py (python main.py or py main.py on Windows)

5. Repository Structure [default]

```
CricScore/
  main.py              entry point
  README.md            full documentation
  requirements-dev.txt pytest
```

```
cricscore/
  match.py          match state: innings, overs, wickets
  scoring.py        rules: runs, extras, run rate, required rate
  storage.py        reading and writing CSV files
  ui.py             prompts and screen output
matches/            saved CSV files (created at runtime)
tests/              pytest tests for the scoring logic
```

Rule of thumb: scoring decides what happens, ui shows it, and storage keeps the player's files.

6. Architecture

Layer	Responsibility	Rules
scoring / match	Runs, extras, wickets, overs, run rate, required rate, innings end	Never prints or reads input. Fully unit-tested
ui	Prompts, input checks and scoreboard display	Contains no scoring rules
storage	Saves ball records and the match summary to CSV	Validates files on load

6.1 Main loop

The app asks for the teams and the number of overs, then repeats one cycle for each ball: read the entry, apply it through the scoring module, redraw the scoreboard and save the ball record. The innings ends after ten wickets, after the last over, or when the target is reached.

6.2 Error handling

An invalid entry, a missing file or a damaged CSV file must never crash the app. The app shows a clear message, asks again, or starts a new file. Corrections to the last ball are supported through an undo command [default].

7. Scoring Rules

Event	Effect [default]
Normal ball	Adds the runs scored and counts as one legal ball
Wide or no-ball	Adds one extra run (plus any runs run); the ball is not counted and is bowled again
Wicket	Adds one to wickets; the ball counts as a legal ball
Over	Complete after six legal balls
Run rate	Total runs divided by overs bowled (balls divided by six)
Required rate	Runs still needed divided by overs remaining

8. Screen Flow (Console)

1. Start screen: new match or exit.
2. Match setup: team names and number of overs.
3. Ball entry: runs, wicket, wide or no-ball, with undo of the last ball.
4. Live scoreboard: score, wickets, overs, run rate and, in the second innings, the target and required rate.
5. Innings break: summary of the first innings and the target.
6. Final scorecard: result, totals and the file name where the match was saved.

9. Data Files

File	Contents

matches/<match_id>_balls.csv	innings, over, ball, runs, extra type, wicket (yes/no)
matches/<match_id>_summary.csv	teams, overs, innings totals and wickets, result

10. Testing and Quality

- Every rule in scoring has a pytest test: an over completes after six legal balls, wides and no-balls do not use a ball, and run rate and required rate are calculated correctly.

11. Documentation Requirements

The README contains a short description, a quick start, how to score a match, an example session, a tour of the folders, how to run the tests, known issues and the roadmap. Functions carry docstrings, and comments explain why, not what.

12. Development Plan

Phase	Content
1	Console scoring of one innings: ball entry, score, wickets, overs, run rate
2	Second innings, target and required rate, CSV saving
3	Input checks, undo, final scorecard, pytest tests, README
4 (stretch)	Graphical interface, player statistics, mobile version

Target dates for each phase: [open]. The first version stays small, the workflow is validated, then the project expands.

13. Risks and Mitigations

Risk	Mitigation
Scoring rules become too complex	Support the basic limited-overs rules first; list advanced rules as non-goals
Wrong entries during a live match	Input checks and an undo command
Scope grows too large	Phased plan; the stretch items are cut first
Match data is lost or damaged	Save after every ball; validate CSV files on load
Users find the console hard to use	Short on-screen prompts and a documented example session

14. Open Questions

1. Should player-wise statistics be included in the first version?
2. Is undo limited to the last ball or does it allow several?
3. Which match formats must be supported (number of overs, number of players)?
4. What will the final file names and folder for saved matches be?

15. Glossary

Over	Six legal balls bowled by one bowler.
Extra	A run added to the score that is not scored off the bat, such as a wide or no-ball.
Run rate	Runs scored per over.
Required rate	Runs per over the batting side still needs to win.
CSV	A plain-text file that stores table data separated by commas.

Project Proposal: Ideation

CricScore: portfolio pages (Section 04 Featured Project and Section 05 90-Second Pitch)

SECTION 04

Featured Project — CricScore

A live match-scoring concept for local and college cricket.

PROBLEM

Scoring local cricket matches on paper is slow and error-prone. Runs, wickets and overs are often miscounted, and players and spectators cannot easily see the live score or run rate.

SOLUTION

CricScore proposes a Python scoring app that records a match ball by ball. The concept includes automatic updates of runs, wickets, overs and run rate, along with a final match summary.

WORKFLOW

1	Teams and number of overs are set up
2	Each ball is entered (runs, wicket, wide, no-ball)
3	Score, wickets and overs update automatically
4	Run rate and required rate are calculated
5	Final scorecard is shown and saved to a CSV file

TECHNICAL DIRECTION

The concept follows a Python, console-first approach and stores match data in CSV files for easy review. It leaves room for a future graphical interface, player statistics and a mobile version.

SECTION 05

CricScore — 90-Second Pitch

Problem → solution → feasibility.

THE PROBLEM

Paper scoring in local cricket is slow, error-prone and hard for spectators to follow live.

THE IDEA

CricScore records a match ball by ball and updates runs, wickets, overs and run rate automatically, so everyone can follow the match.

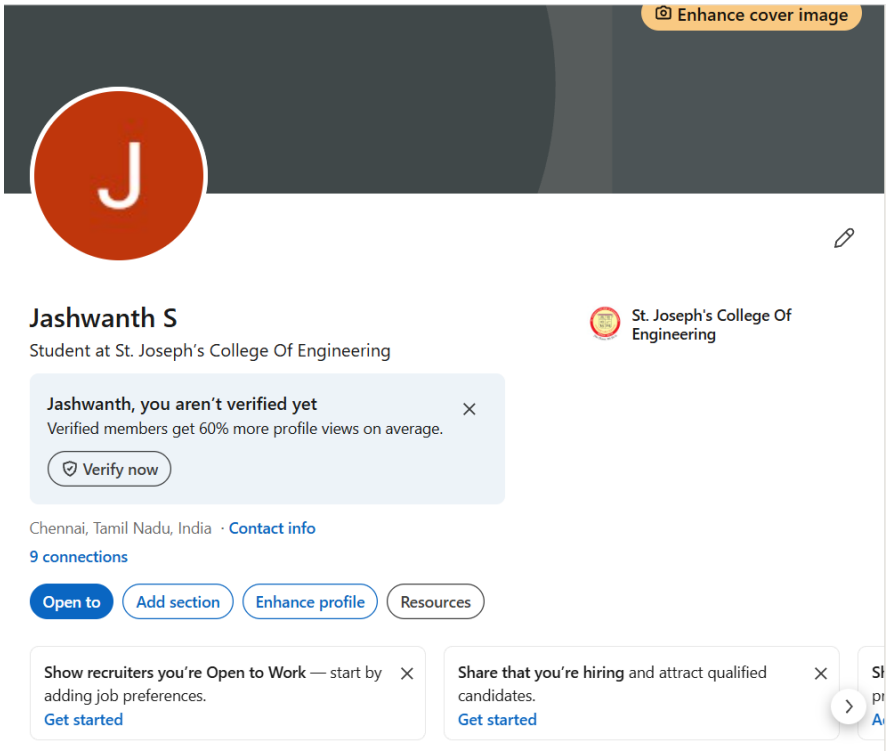
WHY IT IS PRACTICAL

A first version can run in the console using Python and CSV files before adding a graphical interface.

Pilot mindset: start small, validate the workflow, measure useful outcomes, then expand.

LinkedIn and GitHub Profiles

Part A: LinkedIn profile



Name	Jashwanth S
Headline	Student at St. Joseph's College of Engineering
Location	Chennai, Tamil Nadu, India

LinkedIn and GitHub Profiles

Part A: LinkedIn profile (profile text)

Jashwanth S

First-year CSE student | C and Python | Interested in software and application development

About

I'm a first-year Computer Science and Engineering student building a foundation in programming and computational thinking. I focus on learning core concepts clearly, improving problem-solving ability and gradually turning ideas into useful software. My career aim is software and application development.

Focus areas: Programming languages • Application development • Systems and core computing • 3D visuals • AI-assisted coding

Featured Project

CricScore – A live match-scoring concept for local and college cricket. A Python, console-first app that records a match ball by ball, updates runs, wickets, overs and run rate automatically, and saves match data in CSV files. Tech: Python.

Education

B.E. Computer Science and Engineering (CSE-C), 2026–2030, St. Joseph's College of Engineering, Chennai.

Skills

Programming	C, Python
Development	Visual Studio, Git, GitHub
Emerging workflow	AI-assisted coding, vibe coding
Tools explored	Antigravity, OpenCode
Communication	English (developing), Tamil, Hindi (learning)

Career Roadmap

Stage	Focus
1. Foundation	Strengthen C, Python, problem solving and core computer-science concepts.
2. Build	Create small applications and practical projects while using version control.
3. Explore	Develop interest in AI-assisted development, systems and visual computing.
4. Professional growth	Build a strong academic record, portfolio and GitHub presence.

Connect

GitHub	github.com/jashlogin2008-droid
Email	jashlogin80@gmail.com

Part B: GitHub profile



Name	Jashwanth saravanan
Username	jashlogin2008-droid
Bio	I am a quick learner I can do better coding since I am a cs student I am good at python
Location	St Joseph's College of engineering omr chennai
Email	jashlogin80@gmail.com

LinkedIn and GitHub Profiles

Part B: GitHub profile README

Hi there, I'm Jashwanth

CSE Student | C and Python | Software and application development

About Me

- I'm a first-year B.E. Computer Science and Engineering student at St. Joseph's College of Engineering, Chennai.
- I'm building a foundation in programming and computational thinking.
- I work with C and Python and use Git and GitHub for version control.
- I'm interested in AI-assisted coding and have explored tools such as Antigravity and OpenCode.

Tech Stack & Tooling

Languages	C, Python
Development	Visual Studio
Version control	Git, GitHub
Emerging workflow	AI-assisted coding

Featured Project

CricScore (concept) – a live match-scoring app for local and college cricket. Records a match ball by ball and updates runs, wickets, overs and run rate automatically. Python • CSV

Career Roadmap

- **Foundation:** strengthen C, Python, problem solving and core computer-science concepts.
- **Build:** create small applications and practical projects while using version control.
- **Explore:** AI-assisted development, systems and visual computing.
- **Professional growth:** a strong academic record, portfolio and GitHub presence.

Connect With Me

I'm happy to connect with fellow students interested in programming and software development. GitHub: github.com/jashlogin2008-droid | Email: jashlogin80@gmail.com